PHYS4055

Atomic Physics and Quantum Optics

Complete Worked Assignment Solutions

Assignments 1–4

All 12 primary questions, with every subpart

Source-grounded 2026 revision edition

Aim. A step-by-step companion to the 15-page course overview and the 20-example revision book. Every assignment question is unpacked into its physics model, working and final result. Numerical calculations for ^{209}Bi and neutral Na are accompanied by a reproducible, separately versioned Python script.

Assignment	Questions	Main subjects
1: Quantum optics	2	Two-mode squeezed vacuum, thermal marginal, photon pair correlations, beam splitters, coherent-state outputs and photon counting
2: Applications of optics	3	Bell states, partial traces, dense coding, Bell-state detection and BB84 QKD
3: Atomic physics	4	Finite nuclear radius, d^2 terms and NIST spectra, Stark polarisability, Landé factors
4: Atomic applications	3	Hartree–Fock direct/exchange derivation, Zeeman slowing, Na numerical radiative lifetimes

Authority and limitations. The question sheets are included in `source_material/assignments/questions`. Official worked solutions have been released for Assignments **1 and 2**; their methods are cross-checked here. Official solutions for Assignments **3 and 4** were not in the uploaded material: those answers are independently derived. All external atomic constants, data and caveats are documented in `SOURCE_MAP.md`.

First version: 14 November 2026. For numerical work run `python scripts/assignment_numerical_solutions.py` from the repository root.

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Assignment 1

Introduction to quantum optics

1 Question 1: Two-mode squeezed vacuum: thermal reductions and heralding

Source: PHYS4055 2026 Assignment 1, Question 1, source question-sheet page(s) 1.

The input state is

$$|\Psi\rangle_{si} = \sqrt{1-\lambda^2}\sum_{n=0}^{\infty}\lambda^n |n\rangle_s |n\rangle_i , \qquad \lambda = \tanh r.$$

(a) Partial trace and mean occupation

Expand the joint density operator:

$$\rho_{si} = (1-\lambda^2)\sum_{m,n}\lambda^{m+n} |m\rangle_s \langle n|_s \otimes |m\rangle_i \langle n|_i .$$

The identity $\mathrm{Tr}_i(|m\rangle\langle n|) = \delta_{mn}$ makes all off-diagonal signal terms disappear:

$$\boxed{\rho_s = (1-\lambda^2)\sum_{n=0}^{\infty}\lambda^{2n} |n\rangle\langle n|}.$$

The same holds for the idler. Put $q = \lambda^2$ so $p_n = (1-q)q^n$. From $\sum q^n = (1-q)^{-1}$, differentiation gives

$$\bar{n} = (1-q)\sum_{n=0}^{\infty} nq^n = \boxed{\frac{\lambda^2}{1-\lambda^2} = \sinh^2 r}.$$

Since $q = \bar{n}/(1+\bar{n})$, $p_n = \bar{n}^n/(1+\bar{n})^{n+1}$, the single-mode thermal photon-number distribution. The *joint* state is pure while each subsystem is a mixed thermal state.

(b) Cross-correlation

Signal and idler have identical photon numbers in each Schmidt term, so $\langle n_s n_i\rangle = \langle n^2\rangle$. Use $\sum n^2 q^n = q(1+q)/(1-q)^3$:

$$\langle n_s n_i\rangle = \frac{q(1+q)}{(1-q)^2} = \bar{n} + 2\bar{n}^2.$$

Therefore

$$\boxed{g^{(2)}_{si}(0) = \frac{\langle n_s n_i\rangle}{\bar{n}^2} = 2 + \frac{1}{\bar{n}}}.$$

As $r \to 0^+$, pairs become rare but very highly correlated, so the normalised function diverges. At exactly $\bar{n} = 0$, it is undefined (zero denominator).

(c) Herald a single photon

At low gain, $|\Psi\rangle \approx |0,0\rangle + \lambda|1,1\rangle + O(\lambda^2)$. An idler outcome $\langle 1|_i$ leaves an unnormalised signal amplitude proportional to $\lambda|1\rangle_s$. After normalisation,

$$\boxed{\rho_{s|n_i=1} = |1\rangle\langle 1|}.$$

A threshold detector is a good herald only when multi-pair events are negligible; ideal number resolution would isolate the one-pair component exactly.

Exam method / verification. Differentiate the geometric series; do not confuse a local thermal state with an absence of correlations in the joint entangled state.

2 Question 2: Beam splitter on Fock and coherent inputs

Source: PHYS4055 2026 Assignment 1, Question 2, source question-sheet page(s) 1.

The input/output relation is the real rotation

$$\begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} \tau & -\varrho \\ \varrho & \tau \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}.$$

Unitarity means $\tau^2 + \varrho^2 = 1$. The coefficients appearing in the *matrix* are field-amplitude coefficients: the corresponding intensity fractions are τ^2, ϱ^2.

(a) The binomial transformation

The inverse mode relation gives $a^\dagger = \tau c^\dagger + \varrho d^\dagger$. Write the number state with creation operators:

$$|n,0\rangle_{ab} = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle .$$

Expand the commuting creation operators:

$$\frac{1}{\sqrt{n!}} \sum_{k=0}^{n} \binom{n}{k} \tau^k \varrho^{n-k} (c^\dagger)^k (d^\dagger)^{n-k} |0\rangle .$$

Because the normalised number states satisfy $(c^\dagger)^k (d^\dagger)^{n-k} |0\rangle = \sqrt{k!(n-k)!}\, |k, n-k\rangle$, this becomes

$$\boxed{|n,0\rangle_{ab} \longrightarrow \sum_{k=0}^{n} \sqrt{\frac{n!}{k!(n-k)!}}\, \tau^k \varrho^{n-k} |k, n-k\rangle_{cd}}.$$

Normalisation follows from $\sum_k \binom{n}{k} \tau^{2k} \varrho^{2(n-k)} = (\tau^2 + \varrho^2)^n = 1$.

(b) Coherent input plus vacuum

For $c = (a+b)/\sqrt{2},\ d = (a-b)/\sqrt{2}$, write the input as

$$|\alpha, 0\rangle = e^{-|\alpha|^2/2} \exp(\alpha a^\dagger) |0\rangle .$$

The inverse relation is $a^\dagger = (c^\dagger + d^\dagger)/\sqrt{2}$. The two output creation operators commute, so the exponential factors:

$$\begin{aligned} |\Psi_{\text{out}}\rangle &= e^{-|\alpha|^2/2} e^{(\alpha/\sqrt{2})c^\dagger} e^{(\alpha/\sqrt{2})d^\dagger} |0\rangle \\ &= \boxed{\left|\alpha/\sqrt{2}\right\rangle_c \otimes \left|\alpha/\sqrt{2}\right\rangle_d}. \end{aligned}$$

Phase conventions can change the sign of one amplitude without changing output count statistics.

(c) Photodetection

The c-mode coherent amplitude is $\beta = \alpha/\sqrt{2}$. Thus

$$\boxed{P_c(n) = e^{-|\alpha|^2/2} \frac{(|\alpha|^2/2)^n}{n!}}, \qquad \boxed{\langle n_c \rangle = \mathrm{Var}(n_c) = \frac{|\alpha|^2}{2}}.$$

The ideal detector observes *Poissonian* counts.

Exam method / verification. Normalised Fock coefficients need the square roots of factorials; keep amplitude and intensity transmission coefficients distinct.

Assignment 2

Quantum optics applications

3 Question 1: Bell states: orthonormality, reduced state and correlations

Source: PHYS4055 2026 Assignment 2, Question 1, source question-sheet page(s) 1.

We encode $|0\rangle = |H\rangle$, $|1\rangle = |V\rangle$ and define

$$\left|\Phi^{\pm}\right\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad \left|\Psi^{\pm}\right\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

(a) Check every Bell-state overlap

Write $s, t \in \{+1, -1\}$ and $|\Phi_s\rangle = (|00\rangle + s\,|11\rangle)/\sqrt{2}$. Orthonormality of the computational basis gives

$$\langle\Phi_s|\Phi_t\rangle = \frac{1}{2}(1 + st) = \begin{cases} 1 & s = t \\ 0 & s \neq t. \end{cases}$$

Exactly the same reasoning gives $\langle\Psi_s|\Psi_t\rangle = \delta_{st}$. The two families use disjoint computational-basis states, so $\langle\Phi_s|\Psi_t\rangle = 0$. Thus $\boxed{\langle B_i|B_j\rangle = \delta_{ij}}$ for all four Bell states.

(b) Reduced density matrix of photon A

For $|\Phi^+\rangle$,

$$\rho_{AB} = \tfrac{1}{2}(\,|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Use $\mathrm{Tr}_B(|i,j\rangle\langle k,l|) = \delta_{jl}\,|i\rangle\langle k|$. The cross terms vanish and the diagonal terms survive:

$$\boxed{\rho_A = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2} = \frac{1}{2}I}.$$

Therefore $P(H) = P(V) = 1/2$ for measurements on A alone. Its purity $\mathrm{Tr}\,\rho_A^2 = 1/2 < 1$ shows a mixed local state, though ρ_{AB} is pure.

(c) Repeat in the diagonal-polarisation basis

Define $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$, $|A\rangle = (|H\rangle - |V\rangle)/\sqrt{2}$. Hence $|H\rangle = (|D\rangle + |A\rangle)/\sqrt{2}$, $|V\rangle = (|D\rangle - |A\rangle)/\sqrt{2}$. Substitution and cancellation of the two cross terms yields

$$|HH\rangle + |VV\rangle = |DD\rangle + |AA\rangle\,, \qquad \boxed{\left|\Phi^+\right\rangle = \frac{|DD\rangle + |AA\rangle}{\sqrt{2}}}.$$

Both observers using H/V obtain (H,H) or (V,V), each with probability $1/2$. Both using D/A obtain (D,D) or (A,A), again each with probability $1/2$. The local outcomes are random but their *joint* outcomes are perfectly correlated.

Exam method / verification. Distinguish single-photon randomness from two-photon correlation. The partial trace must eliminate off-diagonal terms with unequal B-indices.

4 Question 2: Superdense coding and a passive linear-optical Bell measurement

Source: PHYS4055 2026 Assignment 2, Question 2, source question-sheet page(s) 2–3.

Start with $|\Phi^+\rangle_{AB}$ and act on *Alice's* qubit only. The problem defines $X\,|0\rangle = |1\rangle\,,\ X\,|1\rangle = |0\rangle$, $Z\,|0\rangle = |0\rangle\,,\ Z\,|1\rangle = -\,|1\rangle$.

(a) Calculate the four encoded Bell states

Direct application to each computational term gives

message	gate on A	output state
00	I	$(\|00\rangle + \|11\rangle)/\sqrt{2} = \|\Phi^+\rangle$
01	X	$(\|10\rangle + \|01\rangle)/\sqrt{2} = \|\Psi^+\rangle$
10	Z	$(\|00\rangle - \|11\rangle)/\sqrt{2} = \|\Phi^-\rangle$
11	XZ	$(\|10\rangle - \|01\rangle)/\sqrt{2} = -\|\Psi^-\rangle$

The last global minus phase is unobservable, so the four encodings specify four orthogonal Bell states.

(b) Why two bits can be sent

Before communication, Alice and Bob share one entangled Bell pair. Alice chooses one of four unitaries (2^2 alternatives), sends her *one* photon, and Bob performs an ideal joint Bell-basis measurement. Four orthogonal outputs encode two classical bits. The shared entanglement and the transmitted physical qubit together provide the resource; this is not two bits from an arbitrary isolated one-photon qubit.

(c) Expand the specified beam-splitter measurement

For each polarisation label $\lambda = H, V$,

$$a_\lambda^\dagger \longrightarrow \frac{c_\lambda^\dagger + d_\lambda^\dagger}{\sqrt{2}}, \qquad b_\lambda^\dagger \longrightarrow \frac{c_\lambda^\dagger - d_\lambda^\dagger}{\sqrt{2}}.$$

For two equal polarisations, $a_\lambda^\dagger b_\lambda^\dagger \to [(c_\lambda^\dagger)^2 - (d_\lambda^\dagger)^2]/2$. Therefore

$$\boxed{\left|\Phi^\pm\right\rangle \longrightarrow \frac{(c_H^\dagger)^2 - (d_H^\dagger)^2 \pm (c_V^\dagger)^2 \mp (d_V^\dagger)^2}{2\sqrt{2}} \left|\mathrm{vac}\right\rangle}.$$

For the Ψ states the mixed-polarisation amplitudes interfere:

$$\boxed{\left|\Psi^+\right\rangle \longrightarrow \frac{c_H^\dagger c_V^\dagger - d_H^\dagger d_V^\dagger}{\sqrt{2}} \left|\mathrm{vac}\right\rangle}, \qquad \boxed{\left|\Psi^-\right\rangle \longrightarrow \frac{c_V^\dagger d_H^\dagger - c_H^\dagger d_V^\dagger}{\sqrt{2}} \left|\mathrm{vac}\right\rangle}.$$

A polarising beam splitter and number-resolving detectors at each output distinguish these patterns:

Bell input	Detection signature	Identification
Φ^+	2 photons same port, same pol.	ambiguous
Φ^-	2 photons same port, same pol.	ambiguous
Ψ^+	2 orthogonal pol., same port	unique
Ψ^-	1 photon each port, orthogonal pol.	unique

The relative signs between $\Phi^\pm$ components do not change these photon-count patterns.

(d) Decoding efficiency

With four equiprobable messages, only Ψ^+, Ψ^- are unambiguous:

$$\boxed{P_{\text{unambiguous}} = \frac{2}{4} = \frac{1}{2}}.$$

Thus the standard passive linear-optical analyser alone cannot implement a *deterministic* four-outcome Bell measurement. Extra ancillary resources or different physical interactions may improve the measurement.

Exam method / verification. Bosonic operators for distinct modes commute. Add coherent creation-operator amplitudes before converting into Fock-state probabilities.

5 Question 3: BB84: preparation, sifting, intercept–resend and key distillation

Source: PHYS4055 2026 Assignment 2, Question 3, source question-sheet page(s) 3–4.

(a) Encode bit and basis

The agreed logical encoding is

bit	Z basis	X basis
0	$\|H\rangle$	$\|D\rangle = (\|H\rangle + \|V\rangle)/\sqrt{2}$
1	$\|V\rangle$	$\|A\rangle = (\|H\rangle - \|V\rangle)/\sqrt{2}$

These two bases are mutually unbiased: for example $|\langle H|D\rangle|^2 = 1/2$.

(b) Sift the key and account for loss

Alice and Bob choose each basis with probability 1/2. The probability of matching choices is

$$P(\text{match}) = P(ZZ) + P(XX) = \tfrac{1}{4} + \tfrac{1}{4} = \boxed{\tfrac{1}{2}}.$$

A single photon is detected with probability $\eta_{\text{ch}}\eta_d$, assuming no dark counts or other imperfections:

$$\boxed{\mathbb{E}[N_{\text{det}}] = N\eta_{\text{ch}}\eta_d}, \qquad \boxed{\mathbb{E}[N_{\text{sift}}] = \tfrac{1}{2}N\eta_{\text{ch}}\eta_d}.$$

The sifted-key expectation refers to detected photons for which their announced bases agree.

(c) Derive the error from intercept–resend

Compute the quantum bit error rate *conditioned* on Alice and Bob having used the same basis. Eve chooses that basis with probability 1/2; she then learns and resends the bit without error. With probability 1/2, Eve chooses the conjugate basis. Her outcome is random relative to Alice; when Bob measures in Alice's basis he obtains the wrong bit with probability 1/2. Thus

$$\boxed{Q = \tfrac{1}{2}(0) + \tfrac{1}{2}(\tfrac{1}{2}) = \frac{1}{4} = 25\%.}$$

This is an ideal single-photon intercept–resend attack, not a universal error rate for every attack.

(d.i) Probability of seeing an error

For $m = 20$ independently sampled sifted bits,

$$\boxed{P(\text{at least one error}) = 1 - (1-Q)^m = 1 - \left(\frac{3}{4}\right)^{20} = 0.99683 \text{ (about } 99.68\%).}$$

(d.ii) Why discard public test bits?

The comparison exposes each tested bit on the authenticated public channel. Eve can therefore know their values, so the publicly revealed bits cannot be part of the secret final key.

(d.iii) The two final processing steps

Error correction reconciles any differences between Alice's and Bob's sifted strings, using public parity/syndrome information as necessary. This discussion also leaks some information. **Privacy amplification** compresses the reconciled string with an appropriate hash function to reduce a potential eavesdropper's remaining information to a negligible level.

Exam method / verification. Do not apply the basis-sifting factor twice. For an intercept–resend QBER always condition on a basis match between Alice and Bob.

Assignment 3

Atomic physics

6 Question 1: Finite nuclear size of hydrogen-like ^{209}Bi

Source: PHYS4055 2026 Assignment 3, Question 1, source question-sheet page(s) 1.

Work in atomic units. A point-nucleus Coulomb potential is $V_0 = -Z/r$, while the specified uniform sphere has

$$V_R(r) = \begin{cases} -\frac{Z}{2R}(3 - r^2/R^2) & r < R, \\ -Z/r & r \geq R. \end{cases}$$

Thus the perturbation is

$$\Delta V(r) = Z\left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3}\right) \quad (r < R),$$

and zero for $r \geq R$.

(a) Derive the leading s-state correction

First-order perturbation theory yields

$$\Delta E_{ns}^{(1)} = \langle ns|\Delta V|ns\rangle \simeq 4\pi|\psi_{ns}(0)|^2 \int_0^R r^2 \Delta V(r)\mathrm{d}r.$$

For hydrogenic s-states, $|\psi_{ns}(0)|^2 = Z^3/(\pi n^3)$. Perform the radial integral explicitly:

$$\begin{aligned} \int_0^R r^2 \Delta V(r)\mathrm{d}r &= Z\left[\frac{R^2}{2} - \frac{3R^2}{6} + \frac{R^2}{10}\right] \\ &= \frac{ZR^2}{10}. \end{aligned}$$

Hence

$$\boxed{\Delta E_{ns}^{\mathrm{fns}} = \frac{2Z^4R^2}{5n^3} = \frac{2Z^2}{5n^3}(ZR)^2 \quad \text{(Hartree)}}.$$

It is positive: spreading out the nuclear charge weakens the strongly attractive potential near the origin.

(b) Numerical integrals for ^{209}Bi^{82+}

The nuclear charge is $Z = 83$. With $R_{\mathrm{rms}} = 5.5211\,\mathrm{fm}$, take

$$R = \sqrt{\frac{5}{3}}R_{\mathrm{rms}} = 7.1277\,\mathrm{fm} = 1.34694 \times 10^{-4} a_0.$$

Use normalised hydrogenic $R_{nl}(r)$ and integrate

$$\boxed{\Delta E_{nl} = \int_0^R |R_{nl}(r)|^2 r^2 \Delta V(r)\mathrm{d}r}.$$

The accompanying Python solver performs this quadrature rather than assuming that all l give the same s-state correction:

State	ΔE_{fns} (Ha)	ΔE_{fns} (eV)
$1s$	0.3412153	9.28494
$2p$	9.54585×10^{-8}	2.59756×10^{-6}
$3d$	8.63359×10^{-15}	2.34932×10^{-13}

The leading-order $1s$ formula gives 0.3444055 Ha, approximately 0.94% higher than numerical first-order Schrödinger perturbation theory. The higher angular momenta are strongly suppressed near the nuclear region because $R_{nl} \sim r^l$.

(c) Relativistic comparison

The exact *point-nucleus* Dirac Coulomb energy (excluding rest mass) is

$$E^D_{n\kappa} = \frac{1}{\alpha_{\rm fs}^2}\left[\left(1+\frac{(Z\alpha_{\rm fs})^2}{(n-|\kappa|+\sqrt{\kappa^2-(Z\alpha_{\rm fs})^2})^2}\right)^{-1/2}-1\right]\ \text{Ha}.$$

Compare with $E^{NR}_n = -Z^2/(2n^2)$:

Fine-structure level	$E^D - E^{NR}$ (Ha)	$\Delta E^{NR}_{\rm fns}$ (Ha)
$1s_{1/2}$	-391.87	0.3412
$2p_{1/2}$	-123.80	9.55×10^{-8}
$3d_{3/2}$	-12.31	8.63×10^{-15}

The relativistic shifts are far larger than these *nonrelativistic* finite-nucleus perturbation results. This is not a precision relativistic heavy-ion calculation: $Z\alpha_{\rm fs} \simeq 0.606$, so the real Dirac radial functions have significantly different nuclear-region amplitudes.

Exam method / verification. Numerical source: `scripts/assignment_numerical_solutions.py`. For R_{nl} use $r^2\mathrm{d}r$; for $u = rR_{nl}$ use $\mathrm{d}r$.

7 Question 2: Equivalent d^2 electrons and NIST Ti I spectra

Source: PHYS4055 2026 Assignment 3, Question 2, source question-sheet page(s) 1.

(a) Enumerate the terms

There are $5(2) = 10$ available d-electron spin-orbitals, and $\binom{10}{2} = 45$ antisymmetrised pairs. Coupling $l_1 = l_2 = 2$ permits $L = 0,\ldots,4$; two spins allow $S = 0$ (antisymmetric) or $S = 1$ (symmetric). Orbital exchange symmetry is $(-1)^{2l-L} = (-1)^L$. Consequently even L requires singlet spin and odd L requires triplet spin. Attach $J = |L-S|,\ldots,L+S$:

$$\boxed{{}^1S_0,\ {}^1D_2,\ {}^1G_4,\ {}^3P_{0,1,2},\ {}^3F_{2,3,4}.}$$

The check $1+5+9+(1+3+5)+(5+7+9) = 45$ recovers all permitted microstates.

(b) Hund's ordering

Maximise S, then L. For a less-than-half filled shell, lowest J lies lowest within each term:

$$\boxed{\text{ground } {}^3F_2,\quad {}^3F_2 < {}^3F_3 < {}^3F_4,\quad {}^3P_0 < {}^3P_1 < {}^3P_2.}$$

These rules do not predict an exact ranking between *all different excited terms*.

(c) NIST Ti I data

Neutral Ti has configuration [Ar]$3d^2 4s^2$. The NIST Ti I levels for this configuration are:

Level	cm^{-1}	Level	cm^{-1}
${}^3F_2$	0.000	${}^3F_3$	170.134
${}^3F_4$	386.875	${}^1D_2$	7255.354
${}^3P_0$	8436.617	${}^3P_1$	8492.421
${}^3P_2$	8602.342	${}^1G_4$	12118.394

The ground term and within-multiplet J orders agree with Hund's predictions. The observed ${}^1D_2$ singlet, however, lies *below* the ${}^3P$ triplet, so the simplistic global triplet/singlet ranking fails. The allowed ${}^1S_0$ term is not identified in this short NIST handbook table, and its energy cannot be asserted from these data.

(d) Convert a measured excitation energy

Taking the $^1D_2$ excitation:

$$\boxed{\Delta E = 7255.354\,\mathrm{cm}^{-1} = \frac{7255.354}{219474.63137}\,E_h = 0.0330578\,E_h}.$$

The zero of energy is the atomic ground state, not the ionisation limit.

Exam method / verification. NIST neutral Ti I levels: `physics.nist.gov/PhysRefData/Handbook/Tables/titaniumtable5.htm`. A microstate count is the fastest way to detect missing or Pauli-forbidden terms.

8 Question 3: Hydrogenic static polarisability from perturbation theory

Source: PHYS4055 2026 Assignment 3, Question 3, source question-sheet page(s) 2.

(a) Approximate the $1s$ polarizability

For a uniform electric field in the z direction, $H' = -\mathbf{d}\cdot\boldsymbol{\mathcal{E}} = ez\mathcal{E}$. The $1s$ state has even parity, so $\langle 1s|z|1s\rangle = 0$: its first-order Stark shift vanishes. For a nondegenerate level, second-order perturbation theory gives

$$\Delta E^{(2)} = \mathcal{E}^2 \sum_{m\neq 1} \frac{|\langle m|ez|1\rangle|^2}{E_1 - E_m} = -\tfrac{1}{2}\alpha\mathcal{E}^2.$$

In atomic units, therefore,

$$\alpha = 2\sum_{m\neq 1} \frac{|\langle m|z|1s\rangle|^2}{E_m - E_1}.$$

Approximate every denominator by $E_2 - E_1 = 3Z^2/8$ Hartree. Closure, $\sum_m |m\rangle\langle m| = I$, gives

$$\sum_{m\neq 1} |\langle m|z|1s\rangle|^2 = \langle 1s|z^2|1s\rangle - |\langle 1s|z|1s\rangle|^2 = \langle z^2\rangle.$$

Spherical symmetry implies $\langle z^2\rangle = \langle r^2\rangle/3$. For hydrogenic $1s$, $\langle r^2\rangle = 3/Z^2$, hence

$$\boxed{\alpha_{\mathrm{approx}} = \frac{2(1/Z^2)}{3Z^2/8} = \frac{16}{3Z^4}\,a_0^3.}$$

(b) Hydrogen: numerical comparison

For $Z = 1$, $\alpha_{\mathrm{approx}} = 16/3 = 5.3333\ a_0^3$. With $a_0 = 5.2917721\times 10^{-9}\,\mathrm{cm}$,

$$\boxed{\alpha_{\mathrm{approx}}(H) = 5.3333\,a_0^3 = 7.903\times 10^{-25}\,\mathrm{cm}^3}.$$

This is the standard *polarizability-volume* convention; the SI dipole polarizability is $4\pi\epsilon_0$ times this volume. The exact hydrogen value is $9a_0^3/2 = 4.5a_0^3$. The approximate denominator overestimates it by

$$\boxed{\frac{5.3333 - 4.5}{4.5}\times 100\% = 18.52\%.}$$

(c) Hydrogen-like rubidium

For a nucleus of $Z = 37$ with *one* hydrogenic $1s$ electron:

$$\boxed{\alpha(\mathrm{Rb}^{36+}) \approx \frac{16}{3(37)^4}\,a_0^3 = 2.846\times 10^{-6}\,a_0^3 = 4.217\times 10^{-31}\,\mathrm{cm}^3}.$$

Neutral Rb has outer $5s$ valence electrons which are far more weakly bound and more spatially extended; its ground-state polarisability ($\sim 319\,a_0^3$ as supplied) is about 1.12×10^8 times larger.

Exam method / verification. Parity makes the first-order shift vanish. Closure converts a difficult sum to a ground-state expectation value; the approximate energy denominator is why the result is not exact.

9 Question 4: Landé factors: electronic and nuclear coupling

Source: PHYS4055 2026 Assignment 3, Question 4, source question-sheet page(s) 2.

(a) Derive g_J

Write $\mathbf{J} = \mathbf{L} + \mathbf{S}$, $\boldsymbol{\mu} = -\mu_B(g_L\mathbf{L} + g_S\mathbf{S})$ in dimensionless angular-momentum units, with $g_L = 1,\ g_S = 2$. Only the projection of $\boldsymbol{\mu}$ on $\mathbf{J}$ contributes to the weak-field diagonal energy in a fixed J manifold:

$$g_J = \frac{\langle (g_L\mathbf{L} + g_S\mathbf{S}) \cdot \mathbf{J} \rangle}{J(J+1)}.$$

Using

$$2\mathbf{J}\cdot\mathbf{L} = \mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2, \quad 2\mathbf{J}\cdot\mathbf{S} = \mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2,$$

we find

$$\boxed{g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}} \quad (J > 0).$$

(b) Derive g_F neglecting nuclear magnetism

Now $\mathbf{F} = \mathbf{I} + \mathbf{J}$. The electronic moment lies along $\mathbf{J}$ on average, while $\mathbf{J}$ projects onto $\mathbf{F}$ with factor

$$\frac{\langle \mathbf{J}\cdot\mathbf{F} \rangle}{F(F+1)} = \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}.$$

Thus

$$\boxed{g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}}, \qquad F > 0.$$

This is the electronic contribution only; the nuclear $g_I\mu_N$ term was excluded by the question's stated approximation.

(c) $^2P_{1/2}$ in a weak field

Here $L = 1,\ S = 1/2,\ J = 1/2$ and

$$g_J = 1 + \frac{3/4 + 3/4 - 2}{2(3/4)} = \boxed{\frac{2}{3}}.$$

For $I > 0$, the coupled manifolds are $F_+ = I + \frac{1}{2}$ and $F_- = I - \frac{1}{2}$ (the latter is $F = 0$ for $I = 1/2$). Substituting into the preceding expression gives, for nonzero F,

$$\boxed{g_{I+1/2} = +\frac{2}{3(2I+1)}, \qquad g_{I-1/2} = -\frac{2}{3(2I+1)}}.$$

The weak-field energies are

$$\boxed{\Delta E_{Fm_F} = \mu_B g_F m_F B}, \quad m_F = -F, \ldots, F.$$

The upper/lower hyperfine multiplets fan out into $2F+1$ equally spaced Zeeman levels with *opposite* slopes. For the common $I = 5/2$ example, $F = 3, 2$ with $g_F = +1/9, -1/9$. When $F = 0$, there is only $m_F = 0$ and no first-order electronic Zeeman splitting, so the formula defining $g_{F=0}$ should not be used.

Exam method / verification. Project angular momenta sequentially $L, S \to J$, then $I, J \to F$. In strong fields this coupled-F result no longer applies without state mixing.

Assignment 4

Applications of atomic physics

10 Question 1: Hartree–Fock: direct and exchange Coulomb matrix elements

Source: PHYS4055 2026 Assignment 4, Question 1, source question-sheet page(s) 1.

The task refers specifically to Johnson's closed-shell results, Eqs. (3.64) and (3.66). Here we derive both from the multipole expansion, including the sum over the *occupied* magnetic and spin projections of shell $b = (n_b l_b)$.

1. Radial, orbital and spin separation

In atomic units, let a spin-orbital be

$$\psi_a(\mathbf{r},\sigma) = \frac{P_a(r)}{r} Y_{l_a m_a}(\Omega)\chi_{\mu_a}(\sigma), \qquad \int_0^\infty P_a(r)^2 \mathrm{d}r = 1.$$

For four spin orbitals define

$$g_{abcd} = \int \mathrm{d}x_1 \int \mathrm{d}x_2 \; \frac{\psi_a^*(x_1)\psi_b^*(x_2)\psi_c(x_1)\psi_d(x_2)}{r_{12}}.$$

The Coulomb operator has the spherical expansion

$$\frac{1}{r_{12}} = \sum_{k=0}^{\infty} \sum_{q=-k}^{k} (-1)^q \frac{r_<^k}{r_>^{k+1}} \, C^k_{-q}(\Omega_1) C^k_q(\Omega_2),$$

where $C^k_q = \sqrt{4\pi/(2k+1)} Y_{kq}$. Define the *Slater radial integral*

$$\boxed{R^k(a,b,c,d) = \int_0^\infty \mathrm{d}r_1 \, P_a(r_1) P_c(r_1) \int_0^\infty \mathrm{d}r_2 \, P_b(r_2) P_d(r_2) \frac{r_<^k}{r_>^{k+1}}.}$$

The r^2 Jacobian is already cancelled by the $1/r$ factors in each radial orbital.

2. General angular matrix element

Introduce $A^k_q(a,c) = \langle l_a m_a | C^k_q | l_c m_c \rangle$. Then the multipole expansion separates as

$$g_{abcd} = \delta_{\mu_a \mu_c} \delta_{\mu_b \mu_d} \sum_{kq} (-1)^q R^k(a,b,c,d) A^k_{-q}(a,c) A^k_q(b,d).$$

The Wigner–Eckart theorem gives

$$\langle l_b m_b | C^k_q | l_a m_a \rangle = (-1)^{l_b - m_b} \begin{pmatrix} l_b & k & l_a \\ -m_b & q & m_a \end{pmatrix} \langle l_b \| C^k \| l_a \rangle,$$

where

$$\boxed{\langle l_b \| C^k \| l_a \rangle = (-1)^{l_b} \sqrt{(2l_b+1)(2l_a+1)} \begin{pmatrix} l_b & k & l_a \\ 0 & 0 & 0 \end{pmatrix}.}$$

The rank k must satisfy $|l_a - l_b| \le k \le l_a + l_b$ and $l_a + l_b + k$ even.

3. The direct integral, g_{abab}

In the direct product, each electron keeps its own orbital and spin:

$$g_{abab} = \langle a(1) b(2) | r_{12}^{-1} | a(1) b(2) \rangle.$$

Sum over the two spin values and all $2l_b + 1$ magnetic projections in the *closed* b subshell. The spherical-harmonic addition theorem gives

$$\sum_{m_b=-l_b}^{l_b} |Y_{l_b m_b}(\Omega)|^2 = \frac{2l_b+1}{4\pi}.$$

This summed density is spherically symmetric, so all multipoles $k > 0$ vanish. Only $k = 0$, for which $C_0^0 = 1$, contributes:

$$\boxed{\sum_{m_b,\mu_b} g_{abab} = 2(2l_b + 1)R^0(a, b, a, b).} \qquad \text{(Johnson 3.64)}$$

The factor 2 comes from summing the two possible spin projections of occupied shell b.

4. The exchange integral, g_{abba}

The exchange matrix element swaps the orbital labels on the ket:

$$g_{abba} = \langle a(1)b(2)|r_{12}^{-1}|b(1)a(2)\rangle.$$

Its spin factor is $\delta_{\mu_a\mu_b}\delta_{\mu_b\mu_a}$, so only *same-spin* electrons contribute. The sum over μ_b gives *one*, not two. The angular structure contains the squared transition matrix elements $|\langle l_b m_b|C_q^k|l_a m_a\rangle|^2$. For fixed m_a, 3-j orthogonality states

$$\sum_{m_b,q} \begin{pmatrix} l_b & k & l_a \\ -m_b & q & m_a \end{pmatrix}^2 = \frac{1}{2l_a + 1}.$$

Consequently,

$$\boxed{\sum_{m_b,\mu_b} g_{abba} = \sum_k \frac{|\langle l_b\|C^k\|l_a\rangle|^2}{2l_a + 1} R^k(a, b, b, a).} \qquad \text{(Johnson 3.66)}$$

Only ranks allowed by the triangle and parity selection rules contribute.

5. Why these yield the Hartree–Fock equations

The Coulomb expectation of an antisymmetrised Slater determinant contains the combination

$$\boxed{E_{\text{ee}} = \frac{1}{2} \sum_{a,b \text{ occupied}} \big(g_{abab} - g_{abba}\big).}$$

The first term is the direct, local average electron–electron repulsion. The second is a nonlocal exchange contribution arising from the *antisymmetry* of electron wavefunctions, not a separate classical force. Variational minimisation with orthonormal orbitals gives Hartree–Fock equations with direct and exchange operators. For a single doubly occupied $1s$ subshell, the direct contribution counts both spin occupancies but exchange removes the self-interaction of the occupied spin-orbital.

Exam method / verification. The two target expressions are *shell-summed* matrix elements. Without this sum, neither the factor $2(2l_b + 1)$ nor the reduction to the single $k = 0$ direct term follows. Source: Walter R. Johnson, *Atomic Structure Theory*, chapter 3, Eqs. (3.55)–(3.66).

11 Question 2: Zeeman slower: scattering limit, resonance profile and panels (b)–(e)

Source: PHYS4055 2026 Assignment 4, Question 2, source question-sheet page(s) 1–2.

Consider a beam propagating along $+z$, illuminated by a counter-propagating laser. Let $\Gamma = 1/\tau$ be the excited-state *population decay rate*, $s = I/I_{\text{sat}}$ the saturation parameter and Δ_{eff} the laser detuning in the atom's rest frame.

(a) Maximum scattering rate and design acceleration

The two-level scattering rate is

$$R_{\text{sc}} = \frac{\Gamma}{2} \frac{s}{1 + s + (2\Delta_{\text{eff}}/\Gamma)^2}.$$

Near saturation the maximum is $R_{\max} = \Gamma/2$. Every photon absorbed from the counter-propagating laser changes the atom's directed momentum by approximately $-\hbar k$. Averaging over random spontaneous recoil gives

$$\boxed{F_{\max} = \hbar k\Gamma/2, \quad a_{\max} = \frac{\hbar k\Gamma}{2M}.}$$

A robust real slower typically designs for $a = \eta a_{\max}$, where $0 < \eta < 1$, leaving a margin for power fluctuations, finite linewidth and deviations from exact resonance. Thus, for incident speed v_0, the required ideal length is $L_0 = v_0^2/(2a)$.

(b) Why the optimal field is square-root shaped

Choose a Zeeman shift $\mu' B(z)/\hbar$, where μ' is the effective differential magnetic moment, such that the changing resonance frequency compensates the Doppler shift. With a suitable sign convention, the resonance condition is

$$\Delta_L + kv(z) - \frac{\mu' B(z)}{\hbar} = 0.$$

Therefore $B(z) = B_{\text{offset}} + (\hbar k/\mu')v(z)$ for fixed laser detuning. At constant deceleration,

$$v(z)^2 = v_0^2 - 2az, \qquad \boxed{v(z) = v_0\sqrt{1 - z/L_0}, \quad B(z) = B_{\text{offset}} + B_0\sqrt{1 - z/L_0}},$$

where $B_0 = \hbar k v_0/\mu'$ with the appropriate sign. The square-root profile keeps the photon force close to the chosen constant Ma while the atom slows.

(c) What sets the capture velocity?

Three main limits matter: the *available slower length*, the *maximum sustainable scattering force* and the *magnetic-field range*/laser detuning needed to bring a fast atom into resonance. In the ideal fixed-acceleration model,

$$\boxed{v_{\text{capture}} \lesssim \sqrt{v_f^2 + 2aL}},$$

subject to the entrance-resonance condition

$$|\Delta_L + kv_0 - \mu' B(0)/\hbar| \lesssim \text{the power-broadened linewidth.}$$

A longer slower cannot compensate an insufficient field range or an atom that never enters the laser-coupling resonance window.

(d) Analyse the supplied alternative field plots

The original Assignment 4 page 2 draws $B(z)$ for a reference optimal field (a) and four alternatives (b)–(e). To analyse any alternative, first translate $B(z)$ into the *resonant velocity* $v_{\text{res}}(z) \propto B(z) - B_{\text{offset}}$. Its gradient determines the acceleration needed to stay locked:

$$\boxed{a_{\text{req}}(z) = -v_{\text{res}}(z)\frac{\mathrm{d}v_{\text{res}}}{\mathrm{d}z}.}$$

The atom cannot follow a resonant trajectory requiring $a_{\text{req}} > a_{\max}$, and a *positive* resonant-velocity slope requires the atom to accelerate in the beam direction, opposite to the counter-propagating laser force.

Panel	Observed field shape	Outcome and physical reason
(a)	Reference square root; zero at L_0	Yes in the ideal model: constant allowed deceleration a, reaching $v = 0$ at L_0.
(b)	Shallow straight line; zero near $2L_0$	Can slow to *arbitrarily small* velocity over a longer length. Taking $v_{\text{res}} = v_0(1 - z/(2L_0))$ gives $a_{\text{req}} = a(1 - z/(2L_0)) \leq a$. The mathematical $v = 0$ endpoint is approached asymptotically in time.
(c)	Steep straight line; zero near L_0	At its entrance the required deceleration is $2a$, twice that of reference (a). If (a) was designed at the available force, the atoms fall out of resonance and do not follow the curve to rest.
(d)	A strong down–up–down excursion	The rising-field section corresponds to an *increasing* resonant speed and cannot be tracked by atoms under a purely opposing scattering force. The first steep section can also exceed the force budget. Slowing may resume if the atom is later recaptured, but full stopping is not guaranteed.
(e)	Rapid oscillations before the optimal tail	The ideal resonance velocity oscillates rapidly, demanding alternating acceleration/deceleration. If oscillations exceed the effective linewidth or force response, atoms are detuned and lost; very small slow ripples may be tolerated. Full slowing cannot be assured from the nominal endpoint alone.

A useful additional check for (b): as $v(t) \propto e^{-ct}$, the atom has $v \to 0$ only for $t \to \infty$ within the ideal linearly decreasing resonant-velocity model. Real atoms undergo stochastic photon recoil and exit at a nonzero capture velocity; this makes the distinction between a *mathematical endpoint* and practical capture physically relevant.

Exam method / verification. Always determine whether a field profile lies within the available *force and linewidth* constraints. A plot that reaches $B = 0$ does not on its own guarantee that the atoms follow it or end at rest.

12 Question 3: Neutral sodium: radial eigenstates, E1 rates and lifetimes

Source: PHYS4055 2026 Assignment 4, Question 3, source question-sheet page(s) 1.

This is the computational question. Here we solve a stated effective radial model numerically and show all E1 rate calculations; the corresponding reproducible Python script accompanies the booklet at `scripts/assignment_numerical_solutions.py`.

Step 1: choose the effective central potential

Use the parametric screened potential previously considered in the sodium practical:

$$\boxed{V_b(r) = -\frac{Z}{r} + \frac{Z-\xi}{r}(1-e^{-r/b}), \qquad Z = 11,\ \xi = 1,\ b = 0.4072},$$

with r, b in atomic units. Equivalently $V = -1/r - 10e^{-r/b}/r$. It approaches $-11/r$ close to the nucleus and $-1/r$ in the valence region. It is a *single-electron effective model*, not a full electron–electron Hartree–Fock calculation. For $u_{nl}(r) = rR_{nl}(r)$, solve

$$\boxed{-\frac{1}{2}u''(r) + \left[\frac{l(l+1)}{2r^2} + V_b(r)\right]u(r) = E_{nl}u(r)}, \quad u(0) = u(r_{\max}) = 0, \quad \int_0^{r_{\max}} |u|^2 \mathrm{d}r = 1.$$

Use a finite-difference grid of spacing $h = 0.012\,a_0$ and $r_{\max} = 120\,a_0$. The second derivative uses $u''(r_i) \simeq (u_{i-1} - 2u_i + u_{i+1})/h^2$. The resulting Hamiltonian is tridiagonal:

$$H_{ii} = \frac{1}{h^2} + \frac{l(l+1)}{2r_i^2} + V(r_i), \quad H_{i,i\pm1} = -\frac{1}{2h^2}.$$

Select the correct n, l eigenstates by radial node index $n - l - 1$: $3s$ has two nodes, $4s$ three, $3p$ one, $3d$ zero.

Step 2: numerical energy levels

The calculated energies (relative to ionisation, in Hartree) are

State	l	E_{nl} (Ha)
$3s$	0	-0.1831367
$3p$	1	-0.1102218
$4s$	0	-0.0705131
$3d$	2	-0.0556813

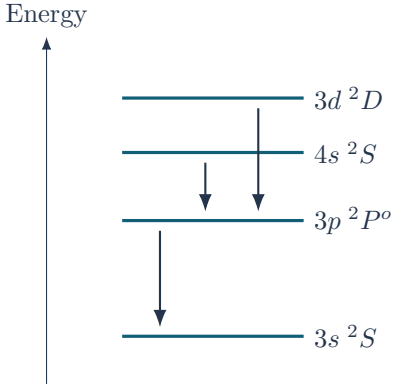
The experimentally compiled sodium excitation energies for comparison are approximately 16956 cm^{-1} for $3p_{1/2}$, 25740 cm^{-1} for $4s$, and 29173 cm^{-1} for $3d$. They are not exactly reproduced because a single spin-independent central potential cannot capture all fine-structure and correlation effects.

Step 3: identify every downward E1 channel

For a single electron, the E1 dipole operator has odd parity and $\Delta l = \pm 1$. With the ordering above, the primary allowed decays are

$$\boxed{3p \to 3s, \qquad 4s \to 3p, \qquad 3d \to 3p.}$$

The direct $4s \to 3s$ and $3d \to 3s$ channels are E1-forbidden ($\Delta l = 0$ and $\Delta l = -2$, respectively). The model level sketch is:



Step 4: evaluate the radial dipole integrals

With $u = rR$, the radial matrix element is

$$D_{if} = \int_0^\infty u_f(r)\, r\, u_i(r) \mathrm{d}r.$$

The numerically computed magnitudes and transition energies are

$i \to f$	$\lvert D_{if}\rvert$ (a_0)	$\omega_{if} = E_i - E_f$ (Ha)
$3p \to 3s$	4.48268	0.0729149
$4s \to 3p$	4.36430	0.0397087
$3d \to 3p$	5.97665	0.0545405

Angular integration over Y_{lm} supplies the E1 weights. For a nonrelativistic spinless orbital with its magnetic substates appropriately averaged, the total downward rate is

$$\boxed{A_{if} = \frac{4\alpha_{\mathrm{fs}}^3}{3t_{\mathrm{au}}}\, \omega_{if}^3\, \frac{l_>}{2l_i + 1}\, |D_{if}|^2,}$$

where $l_> = \max(l_i, l_f)$, $t_{\mathrm{au}} = 2.4188843266 \times 10^{-17}$ s. The factors $l_>/(2l_i + 1)$ are 1/3 for $p \to s$, 1 for $s \to p$ and 2/5 for $d \to p$.

Step 5: sum rates and calculate lifetimes

The excited-state lifetime is $\tau_i = [\sum_{f<i} A_{if}]^{-1}$. With the specified model, each upper state has one primary orbital E1 channel:

i	Calculated A_i (s^{-1})	τ_i (ns)
$3p$	5.562×10^7	17.979
$4s$	2.545×10^7	39.147
$3d$	4.965×10^7	20.139

These are independently computed predictions, not values fitted to the reference lifetimes.

Step 6: compare with published atomic data

Published nonrelativistic/relativistic atomic-structure calculations for the Na-like sequence at $Z = 11$ report approximately 16.48 ns for $3p$, 38.25 ns for $4s$, and 19.67 ns for $3d$; the J-resolved values vary slightly. Our percentage differences from these *theoretical reference* results are:

	this model (ns)	ref. theory (ns)	difference
$3p$	17.979	16.48	+9.10%
$4s$	39.147	38.25	+2.34%
$3d$	20.139	19.67	+2.39%

A separately published direct measurement of Na I $3p$ reports 17.3 ± 1.0 ns, consistent with our predicted 17.98 ns at that uncertainty. The $4s$ and $3d$ benchmark values in the table are *calculations*, not claimed direct measurements; a precise experimental comparison should use a cited measured dataset for each fine-structure component.

Step 7: limitations

This simple model ignores core polarisation, electron correlation beyond the fitted potential, spin–orbit fine structure and configuration mixing. Its chosen b parameter controls the radial overlaps; different fitted potentials change τ. Higher multipoles, collisional quenching, cascades and branching into more levels may matter outside the minimal model. Numerical convergence should be checked by decreasing h and increasing $r_{\max}$.

Exam method / verification. Reproduce every number using `python scripts/assignment_numerical_solutions.py`. References: Froese Fischer *et al.*, Atomic Data and Nuclear Data Tables **92** (2006), Na-like $Z = 11$ lifetimes; NIST Na I energy-level table; 1970 JOSA Na I $3p$ lifetime measurement.

Question-type method finder

Question	First step to remember
A1 Q1	Expand $\|\Psi\rangle\langle\Psi\|$, trace idler with δ_{mn}, differentiate geometric series
A1 Q2	Write Fock state using creation operators, invert the optical mode matrix, apply binomial theorem
A2 Q1	Check Bell overlaps, trace one subsystem, transform both photons to diagonal basis
A2 Q2	Apply local Pauli gates, transform *both* photonic creation operators and classify detector signatures
A2 Q3	Sift based on random basis match, condition intercept–resend error on Alice/Bob matching bases
A3 Q1	Form $\Delta V = V_{\text{finite}} - V_{\text{point}}$, integrate over nuclear radius and compare Dirac corrections
A3 Q2	Enforce orbital/spin exchange antisymmetry, count 45 states; check NIST level order
A3 Q3	Use second-order Stark expression, symmetry and closure; convert a_0^3 to polarizability volume
A3 Q4	Project $L, S \to J$, then $I, J \to F$; split in m_F in a weak magnetic field
A4 Q1	Separate radial/angular/spin; sum m_b, μ_b before writing Johnson's 3.64 and 3.66
A4 Q2	Max scattering force sets $a_{\max}$; use $v\mathrm{d}v/\mathrm{d}z$ to test a field profile
A4 Q3	Solve the radial tridiagonal Hamiltonian; calculate E1 angular factors and radial dipoles; sum rates then invert to lifetime

Appendix: reproducible numerical code

The following source file is also committed separately for execution and modification. It calculates the finite-nuclear-size corrections, Dirac comparison, and sodium radial energies and lifetimes.

```python
#!/usr/bin/env python3
"""PHYS4055 reproducible numeric assignment answers (2026).

Assignment 3 Q1: hydrogenic 209Bi finite-nuclear-size shifts and point-nucleus Dirac
comparisons. Assignment 4 Q3: nonrelativistic radial effective-potential model
of neutral Na, E1 dipole amplitudes, and 3p / 4s / 3d lifetimes.

Run: python -m pip install numpy scipy
    python scripts/assignment_numerical_solutions.py
All energies and radial distances are in atomic units unless stated.
"""
import math
import numpy as np
from scipy.integrate import quad
from scipy.linalg import eigh_tridiagonal
from scipy.special import eval_genlaguerre

ALPHA = 1 / 137.035999084
TAU_AU_S = 2.4188843265857e-17
BOHR_FM = 52917.7210903
EV_PER_HARTREE = 27.211386245988
RMS_BI209_FM = 5.5211 # Angeli & Marinova (2013), rms nuclear charge radius.


def hydrogen_radial(r: float, n: int, l: int, Z: int) -> float:
    rho = 2.0 * Z * r / n
    prefactor = (2 * Z / n)**1.5 * math.sqrt(
        math.factorial(n-l-1) / (2 * n * math.factorial(n+l))
    )
    return prefactor * math.exp(-rho/2) * rho**l * eval_genlaguerre(n-l-1, 2*l+1, rho)


def fns_deltaV(r: float, Z: int, R: float) -> float:
    if r >= R: return 0.0
    return Z*(1/r - 3/(2*R) + r*r/(2*R**3))


def hydrogenic_bi_shifts():
    Z = 83
    R = math.sqrt(5/3)*RMS_BI209_FM/BOHR_FM
    print("ASSIGNMENT 3 Q1. 209Bi nuclear-size shifts")
    print(f"Uniform nuclear radius = {R:.10g} Bohr = {R*BOHR_FM:.7f} fm")
    print("State finite-size (Ha) finite-size (eV) Dirac minus Schrodinger (Ha)")
    for n,l,label,kappa in [(1,0,"1s",-1),(2,1,"2p1/2",+1),
                            (3,2,"3d3/2",+2),(3,2,"3d5/2",-3)]:
        integral,_ = quad(
            lambda x: abs(hydrogen_radial(R*x,n,l,Z))**2
                * (R*x)**2 * fns_deltaV(R*x,Z,R) * R,
            1.0e-12, 1.0, epsabs=1e-20, epsrel=1e-12, limit=200)
        mc2 = 1/ALPHA**2
        gam = math.sqrt(kappa*kappa-(Z*ALPHA)**2)
        E_dirac = mc2*((1+(Z*ALPHA)**2/(n-abs(kappa)+gam)**2)**-0.5-1)
        E_sch = -Z**2/(2*n*n)
        print(f"{label:7} {integral:17.10e} {integral*EV_PER_HARTREE:18.10e} {E_dirac-E_sch:22.9f}")
    leading = 2*Z**4*R**2/5
    print("1s small-R leading shift:",f"{leading:.10f} Hartree")


def na_states(step=0.012, r_max=120.0):
    """Radial finite-difference solver with Dirichlet boundaries at r=0, r=rmax.

    u(r)=r R_nl(r), integral |u|^2 dr=1.
    Effective potential V=-Z/r+(Z-xi)(1-exp(-r/b))/r, Z=11, xi=1, b=.4072.
    Retain states labelled by n and l, using the radial node index n-l-1.
    """
    Z,xi,b = 11,1,0.4072
    r = np.arange(1,int(r_max/step)+1,dtype=float)*step
    V = -xi/r - (Z-xi)*np.exp(-r/b)/r
    wanted = {"3s":(0,2), "4s":(0,3), "3p":(1,1), "3d":(2,0)}
    states={}
    for l in [0,1,2]:
        diagonal = 1/step**2 + V + l*(l+1)/(2*r**2)
        offdiagonal = np.full(len(r)-1,-1/(2*step**2))
        energies, vectors=eigh_tridiagonal(
            diagonal,offdiagonal,select="i",select_range=(0,4))
        for label,(wanted_l,k) in wanted.items():
            if l==wanted_l:
                u=vectors[:,k]/math.sqrt(step)
                states[label] = (float(energies[k]),u)
    return r,states


def sodium_lifetimes(step=0.012, r_max=120.0):
    r,states=na_states(step,r_max)
    print("\nASSIGNMENT 4 Q3. Na effective-potential E1 lifetimes")
    print("Radius step =",step,"a0; r_max =",r_max,"a0")
    for name in ("3s","3p","4s","3d"):
```

```
        print(f"{name:4} E={states[name][0]:.9f} Ha")
    print("Transition radial dipole(a0) omega(Ha) A(s^-1) lifetime(ns)")
    for src,dst,factor in [("3p","3s",1/3),("4s","3p",1),("3d","3p",2/5)]:
        E_i,u_i=states[src];E_f,u_f=states[dst]
        radial=float(np.trapezoid(u_i*u_f*r,dx=step))
        omega=E_i-E_f
        assert omega>0, "Final state must have lower energy"
        A=(4/3)*ALPHA**3*(omega**3)*factor*radial**2/TAU_AU_S
        tau_ns=1e9/A
        print(f"{src}->{dst:2} {radial:18.7f} {omega:12.8f} {A:11.4e} {tau_ns:14.5f}")
    print("Reference theoretical lifetimes: Na 3p ~=16.48 ns, 4s ~=38.25 ns,")
    print("3d ~=19.7 ns (Froese Fischer et al., ADND Tables 92 [2006], Z=11).")
    print("Na 3p historical direct measurement: 17.3 +/- 1.0 ns")
    print("(JOSA 60, 1199 [1970]: Atomic-Lifetime Measurements for Neutral")
    print("and Ionized Sodium and Potassium; experimental Na I 3p result).")


if __name__=="__main__":
    hydrogenic_bi_shifts()
    sodium_lifetimes()
    print("\nGrid-convergence check:")
    for h in [0.024, 0.016,0.012,0.010]:
        print(f"--- h = {h} ---")
        sodium_lifetimes(h, 120.0)
```